# 1.2 Finite Fields from Modular Arithmetic

Every field of Section 1.1 is infinite. *Number Systems: From* $\mathbb{N}$ *to* $\mathbb{R}$ has already met finite ones: it proved that the integers modulo a prime $p$ form a field (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.2). This section restates that result in the language of Definition 1.1.1, gives these fields a name of their own, $\mathbb{F}_p$, and computes in them: inverses, linear equations, and the elimination method of Section 1.1, now run modulo 5. Finite fields are not a curiosity; coding theory and computer algebra do linear algebra over them every day.

## 1.2.1 Arithmetic modulo n, recalled

Fix an integer $n \geq 1$. The ring $\mathbb{Z}/n\mathbb{Z}$ of Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2 consists of the classes $[a]$ of integers $a$ under congruence modulo $n$, and two classes are equal exactly when their representatives differ by a multiple of $n$:

$$[a] = [b] \iff n \mid a - b.$$

This is the class test of *Number Systems* (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (a), read with Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2). Sums and products are computed on representatives. The first example collects what is known about this arithmetic, in the language of Definition 1.1.1.

**Example. Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$**

Let $n \geq 1$, and define $[a] + [b] := [a + b]$ and $[a][b] := [ab]$ for $a, b \in \mathbb{Z}$. Show that these operations are well defined, that they satisfy every clause of Definition 1.1.1 except possibly (F8) and the condition $1 \neq 0$ in (F7), and that $[1] \neq [0]$ holds exactly when $n \geq 2$. Then write out the addition and multiplication tables for $n = 2$.

*Solution.* All of this was proved in *Number Systems*, and we collect it.

*Well-definedness.* The formulas use representatives, so the answer must not depend on which representatives are chosen. For the ideal $n\mathbb{Z}$ of $\mathbb{Z}$ this is Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.1 (b), and $\mathbb{Z}/n\mathbb{Z}$ is defined as that quotient (Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2).

*The clauses.* By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (a), $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with zero $[0]$ and one $[1]$: clauses (F1)–(F6) and (F9) hold, and so does (F7) without the condition $1 \neq 0$. The negative of $[a]$ is $[-a]$, since $[a] + [-a] = [a + (-a)] = [0]$.

*The condition* $1 \neq 0$. By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (c), $[1] \neq [0]$ if and only if $n \geq 2$. For $n = 1$ every class equals $[0]$, and $\mathbb{Z}/1\mathbb{Z}$ is the zero ring.

*Size.* By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (b), each class is $[r]$ for exactly one $r \in \{0, 1, \ldots, n-1\}$, the remainder of any representative on division by $n$. So $\mathbb{Z}/n\mathbb{Z}$ has exactly $n$ elements.

*The tables for* $n = 2$. Write 0 and 1 for $[0]$ and $[1]$. Since $1 + 1 = 2$ and $2 - 0 = 2 \cdot 1$, the

class test gives $[1]+[1]=[2]=[0]$; the other entries are read off directly:

| $+$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 1 | 0 |

| $\cdot$ | 0 | 1 |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 0 | 1 |

Here (F8) holds as well: the only non-zero element is 1, and $1 \cdot 1 = 1$. So $\mathbb{F}_2 \coloneqq \mathbb{Z}/2\mathbb{Z}$ is a field. In it $1+1=0$, and therefore $-1=1$.

The tables of $\mathbb{F}_2$ are those of the two-element field of Number Systems: From \(\nN\) to \(\nR\), Example (A field with two elements that cannot be ordered), with 0 and 1 read as classes. The one surprise is $1+1=0$: it says that 2 is even, and nothing more. It is also the field promised by the warning on order in Section 1.1, where "$1+1\neq 0$" fails.

## 1.2.2 Which moduli give fields

For $n \geq 2$, whether $\mathbb{Z}/n\mathbb{Z}$ is a field now comes down to one clause, (F8): does every non-zero class have an inverse? An inverse of $[a]$ is a class $[x]$ with $[ax]=[1]$, that is, an integer $x$ with $n \mid ax-1$, that is, integers $x, y$ with $ax+ny=1$. Such a combination is what Bézout's theorem produces. The linear-algebra volumes cite it in the following special form.

**Lemma 1.2.1. Bézout's Lemma for Integers**

Let $a, n \in \mathbb{Z}$ with $n \geq 1$, and suppose that the only positive integer dividing both $a$ and $n$ is 1. Then there are $x, y \in \mathbb{Z}$ with

$$ax + ny = 1.$$

*Proof.* Since $n \geq 1$, $(a,n) \neq (0,0)$, so $g \coloneqq \gcd(a,n)$ is a common divisor of $a$ and $n$ with $g \geq 1$ (Number Systems: From \(\nN\) to \(\nR\), Definition 3.2.2 and the well-definedness paragraph after it). By hypothesis, $g=1$. So $a$ and $n$ are coprime, and Number Systems: From \(\nN\) to \(\nR\), Corollary 3.2.2 gives $x, y \in \mathbb{Z}$ with $ax+ny=1$. ∎

The general statement, for any pair of integers not both zero and with $\gcd(a,n)$ in place of 1, is Number Systems: From \(\nN\) to \(\nR\), Theorem 3.2.1; the Euclidean algorithm computes the $x$ and $y$. We keep the special case under its own name because the linear-algebra volumes cite it in this form.

**Theorem 1.2.2. $\mathbb{Z}/n\mathbb{Z}$ Is a Field Exactly for Prime $n$**

Let $n \geq 2$ be an integer. Then $\mathbb{Z}/n\mathbb{Z}$ is a field (Definition 1.1.1) if and only if $n$ is prime.

*Idea.* By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), only (F8) is in question. If $n=p$ is prime and $[a]\neq[0]$, then $p \nmid a$, so the only positive common divisor of $a$ and $p$ is 1, and Lemma 1.2.1 hands over $ax+py=1$, that is, $[a][x]=[1]$. If instead $n=rs$ with $1<r,s<n$, then $[r][s]=[n]=[0]$ with both factors non-zero, which Theorem 1.1.1 (f) forbids in a field. *Number Systems* ran this argument, with Number Systems: From \(\nN\)

to \(\nR\), Theorem 3.5.1 in the role of the lemma.

*Proof.* By Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.2, $\mathbb{Z}/n\mathbb{Z}$ is a field in the sense of Number Systems: From \(\nN\) to \(\nR\), Definition 2.3.1 if and only if $n$ is prime. By the remark after Definition 1.1.1, a field in that sense is the same as a field in the sense of Definition 1.1.1. This proves the theorem.

■

So for each prime $p$ there is a field with exactly $p$ elements, and for each composite $n \geq 4$ the ring $\mathbb{Z}/n\mathbb{Z}$ is not a field. The fields deserve a short name.

## 1.2.3 The fields with p elements

**Definition 1.2.1. The Field $\mathbb{F}_p$**

Let $p$ be a prime. The field $\mathbb{F}_p$ is $\mathbb{Z}/p\mathbb{Z}$, with the operations of Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$). It has exactly $p$ elements (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (b)), which we write $0, 1, \ldots, p-1$, without brackets: the number $r \in \{0, \ldots, p-1\}$ stands for the class $[r]$. For integers $a, b$, "$a = b$ in $\mathbb{F}_p$" means $[a] = [b]$, that is, $p \mid a - b$.

So $\mathbb{F}_2$, $\mathbb{F}_3$, $\mathbb{F}_5$, $\mathbb{F}_7$, $\mathbb{F}_{11}$, ... are fields, by Theorem 1.2.2, while $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$ and $\mathbb{Z}/9\mathbb{Z}$ are not. To decide that a given $p$ is prime, the remark after Number Systems: From \(\nN\) to \(\nR\), Lemma 3.3.4 says it is enough to check that no $d \geq 2$ with $d \cdot d \leq p$ divides $p$.

Computing in $\mathbb{F}_p$ is computing with integers and replacing the answer by its remainder on division by $p$. The words "in $\mathbb{F}_p$" must be said whenever an equation would be false in $\mathbb{Z}$: "$3 \cdot 5 = 1$ in $\mathbb{F}_7$" is true, because $15 - 1 = 2 \cdot 7$; "$3 \cdot 5 = 1$" alone is false. Every result of Section 1.1 now holds in $\mathbb{F}_p$. For instance, by Corollary 1.1.2 the only elements of $\mathbb{F}_p$ whose square is 1 are 1 and $-1 = p - 1$.

*Quick check.* Find the inverse of 3 in $\mathbb{F}_7$.

*Solution.* The inverse is the $x \in \{1, \ldots, 6\}$ with $7 \mid 3x - 1$. Testing the candidates in turn, the products $3, 6, 9, 12, 15$ leave remainders $3, 6, 2, 5, 1$, and $3 \cdot 5 = 15 = 2 \cdot 7 + 1$. So $3^{-1} = 5$ in $\mathbb{F}_7$, and by Theorem 1.1.1 (c) it is the only inverse.

For small $p$, trial is fast. For larger $p$ the Euclidean algorithm of *Number Systems* finds the inverse directly: if $ax + py = 1$, then $\gcd(a, p) = 1$ by Number Systems: From \(\nN\) to \(\nR\), Corollary 3.2.2, and Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.1 gives $[a]^{-1} = [x]$.

**Example. Inverses in $\mathbb{F}_7$ and in $\mathbb{F}_{101}$**

(i) List the inverses of the non-zero elements of $\mathbb{F}_7$. (ii) Find the inverse of 17 in $\mathbb{F}_{101}$.

*Solution.*

(i) In $\mathbb{F}_7$:

$$1 \cdot 1 = 1, \quad 2 \cdot 4 = 8 = 1, \quad 3 \cdot 5 = 15 = 1, \quad 6 \cdot 6 = 36 = 1,$$

since $8-1=7$, $15-1=2\cdot 7$ and $36-1=5\cdot 7$. So $1^{-1}=1$, $2^{-1}=4$, $4^{-1}=2$, $3^{-1}=5$, $5^{-1}=3$, $6^{-1}=6$, using (F6) for the reversed pairs. Each element of $\mathbb{F}_7^{\times} := \mathbb{F}_7 \setminus \{0\}$ appears exactly once as an inverse, as it must: the inverse of $a^{-1}$ is $a$ (Theorem 1.1.1 (g)), so two elements with the same inverse are equal. The elements 1 and $6=-1$ are their own inverses, and by Corollary 1.1.2 no others are.

(ii) First, 101 is prime, by the criterion recalled after Definition 1.2.1: no $d$ with $2 \le d \le 10$ divides it (the remainders of 101 on division by $2, 3, \ldots, 10$ are $1, 2, 1, 1, 5, 3, 5, 2, 1$), and $11 \cdot 11 = 121 > 101$. Run the Euclidean algorithm on $(101, 17)$:

$$101 = 5 \cdot 17 + 16, \qquad 17 = 1 \cdot 16 + 1.$$

Back-substitute from the last line:

$$1 = 17 - 16 = 17 - (101 - 5 \cdot 17) = 6 \cdot 17 - 101.$$

So $17 \cdot 6 + 101 \cdot (-1) = 1$, hence $\gcd(17, 101) = 1$ (Number Systems: From \(\nN\) to \(\nR\), Corollary 3.2.2), and Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.1 gives $17^{-1} = 6$ in $\mathbb{F}_{101}$. Check: $17 \cdot 6 = 102 = 101 + 1$.

Since $\mathbb{F}_p$ is a field, each result of Section 1.1 about equations holds there word for word; what changes is how the numbers are computed. An elimination of the same kind as Section 1.1's opening one, done over $\mathbb{F}_5$, looks like this.

**Example. Elimination over $\mathbb{F}_5$**

Solve the system $2x + y = 3$, $3x + 2y = 4$ over $\mathbb{F}_5$.

*Solution.* In $\mathbb{F}_5$ the inverses are $1^{-1}=1$, $2^{-1}=3$, $3^{-1}=2$ and $4^{-1}=4$, since $2\cdot 3 = 6 = 5+1$ and $4 \cdot 4 = 16 = 3 \cdot 5 + 1$.

*By elimination.* Scale the first equation by $2^{-1}=3$. In $\mathbb{F}_5$, $3\cdot 2 = 1$, $3\cdot 1 = 3$ and $3\cdot 3 = 9 = 4$, so the scaled equation reads $x + 3y = 4$. Taking 3 times it away from the second equation removes $x$ and leaves

$$(2-9)y = 4 - 12, \quad \text{that is,} \quad 3y = 2 \ \text{ in } \mathbb{F}_5,$$

since $-7 = -10 + 3$ and $-8 = -10 + 2$. Multiply by $3^{-1} = 2$: $y = 4$. Then $x = 4 - 3y = 4 - 12 = -8 = 2$ in $\mathbb{F}_5$.

*By the formula.* With $a=2$, $b=1$, $c=3$, $d=2$, $e=3$, $f=4$, the coefficient $ad - bc =$

$4 - 3 = 1$ is non-zero, so Proposition 1.1.3 (b) applies:

$$x = \frac{3 \cdot 2 - 1 \cdot 4}{1} = 2, \qquad y = \frac{2 \cdot 4 - 3 \cdot 3}{1} = -1 = 4 \text{ in } \mathbb{F}_5.$$

The two methods agree. Check: $2 \cdot 2 + 4 = 8 = 3$ and $3 \cdot 2 + 2 \cdot 4 = 14 = 4$ in $\mathbb{F}_5$. By Proposition 1.1.3 (b), $(x, y) = (2, 4)$ is the only solution.

*Quick check.* Which elements $x \in \mathbb{F}_{11}$ satisfy $x^2 = 1$?

*Solution.* Exactly 1 and 10. By Corollary 1.1.2, $x^2 = 1$ if and only if $x = 1$ or $x = -1$, and $-1 = 10$ in $\mathbb{F}_{11}$, since $10 - (-1) = 11$. Check: $10^2 = 100 = 9 \cdot 11 + 1$.

*Warning.* A non-zero class of $\mathbb{Z}/n\mathbb{Z}$ need not have an inverse. In $\mathbb{Z}/6\mathbb{Z}$ the equation $[2]x = [1]$ has **no** solution, since $[2]x$ is always one of $[0], [2], [4]$; and $[2]x = [4]$ has **two** solutions, $x = [2]$ and $x = [5]$ (as $10 - 4 = 6$). So division by $[a]$ modulo $n$ is allowed when $n$ is prime and $[a] \neq [0]$, or more generally when $\gcd(a, n) = 1$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.5.1), and not otherwise.

*Remark* (No order on $\mathbb{F}_p$). For $n \geq 2$, no relation $<$ makes $\mathbb{Z}/n\mathbb{Z}$ an ordered commutative ring (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (e)). So the inequalities of *Number Systems* have no meaning in $\mathbb{F}_p$, as the warning on order in Section 1.1 said. The representatives $0, 1, \ldots, p - 1$ are ordered as integers, but that order is not compatible with the arithmetic: $0 < p - 1$ as integers, and adding 1 to both sides would give $1 < 0$, since $(p - 1) + 1 = 0$ in $\mathbb{F}_p$.

## 1.2.4 Exercises

### 1.2.4.1 A. Check your understanding

**Exercise. A1**

(a) True or false: $\mathbb{Z}/8\mathbb{Z}$ is a field. Justify your answer.

(b) List the inverses of $1, 2, 3, 4$ in $\mathbb{F}_5$.

(c) Compute $4 \cdot 4 + 3$ in $\mathbb{F}_5$.

(d) True or false: $x^2 = x$ for every $x \in \mathbb{F}_2$.

(e) How many elements does $\mathbb{F}_{13}$ have, and is $13 = 0$ in $\mathbb{F}_{13}$?

*Solution.*

(a) False. $8 = 2 \cdot 4$ is not prime, so Theorem 1.2.2 applies; concretely $[2][4] = [8] = [0]$ with $[2] \neq [0]$ and $[4] \neq [0]$, which Theorem 1.1.1 (f) forbids in a field.

(b) $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, $4^{-1} = 4$, as in Example (Elimination over $\mathbb{F}_5$).

(c) $4 \cdot 4 + 3 = 19 = 3 \cdot 5 + 4$, so the answer is 4.

(d) True: $0 \cdot 0 = 0$ and $1 \cdot 1 = 1$ in $\mathbb{F}_2$. (This agrees with Exercise (B1) (c): the solutions of $x^2 = x$ in any field are 0 and 1, and $\mathbb{F}_2$ has no other elements.)

(e) 13 is prime (neither 2 nor 3 divides it, and $4 \cdot 4 = 16 > 13$), so $\mathbb{F}_{13}$ is defined and has 13 elements. And $13 = 0$ in $\mathbb{F}_{13}$, since $13 \mid 13 - 0$.

#### 1.2.4.2 B. Practice

**Exercise. B1**

(a) Solve $3x + 5 = 2$ in $\mathbb{F}_7$.

(b) Solve the system $x + 2y = 3$, $3x + y = 1$ over $\mathbb{F}_7$.

*Solution.*

(a) Adding $-5$ to both sides gives $3x = 2 - 5 = -3 = 4$ in $\mathbb{F}_7$. Since $3 \cdot 5 = 15 = 1$ in $\mathbb{F}_7$, $3^{-1} = 5$, and Proposition 1.1.3 (a) gives $x = 5{\cdot}4 = 20 = 6$. Check: $3{\cdot}6{+}5 = 23 = 3{\cdot}7{+}2$.

(b) Here $a = 1$, $b = 2$, $c = 3$, $d = 1$, $e = 3$, $f = 1$, and $ad - bc = 1 - 6 = -5 = 2 \neq 0$ in $\mathbb{F}_7$. Since $2 \cdot 4 = 8 = 1$, $2^{-1} = 4$. By Proposition 1.1.3 (b),

$$x = (3 \cdot 1 - 2 \cdot 1) \cdot 4 = 4, \qquad y = (1 \cdot 1 - 3 \cdot 3) \cdot 4 = -32 = 3,$$

since $-32 = -5 \cdot 7 + 3$. Check: $4 + 2 \cdot 3 = 10 = 3$ and $3 \cdot 4 + 3 = 15 = 1$ in $\mathbb{F}_7$. Hence $(x, y) = (4, 3)$, and it is the only solution.

**Exercise. B2: Which Are Fields?**

Determine which of $\mathbb{Z}/9\mathbb{Z}$, $\mathbb{F}_5$, $\mathbb{Z}/1\mathbb{Z}$ and $\mathbb{Z}/91\mathbb{Z}$ are fields. Justify your answer.

*Solution.* $\mathbb{Z}/9\mathbb{Z}$: **not a field**, since $[3][3] = [9] = [0]$ with $[3] \neq [0]$, against Theorem 1.1.1 (f). $\mathbb{F}_5$: **a field**, by Theorem 1.2.2, since 5 is prime. $\mathbb{Z}/1\mathbb{Z}$: **not a field**, since $[1] = [0]$ there (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)), against the condition $1 \neq 0$ of (F7). $\mathbb{Z}/91\mathbb{Z}$: **not a field**, since $91 = 7 \cdot 13$ is not prime (Theorem 1.2.2); concretely $[7][13] = [91] = [0]$.

**Exercise. B3**

Find all solutions of $x^2 = 2$ in $\mathbb{F}_7$, and all solutions of $x^2 = 2$ in $\mathbb{F}_5$.

*Solution.* In $\mathbb{F}_7$, $3^2 = 9 = 2$. By Corollary 1.1.2, $x^2 = 2 = 3^2$ holds exactly for $x = 3$ and $x = -3 = 4$. Check: $4^2 = 16 = 2 \cdot 7 + 2$.

In $\mathbb{F}_5$ the squares of $0, 1, 2, 3, 4$ are $0, 1, 4, 9 = 4, 16 = 1$. None is 2, so $x^2 = 2$ has no solution in $\mathbb{F}_5$.

#### 1.2.4.3 C. Going deeper

**Exercise. C1: A Field with Four Elements**

Let $F$ be the set of formal expressions $a + b\alpha$ with $a, b \in \mathbb{F}_2$: four symbols 0, 1, $\alpha$ and $\beta := 1 + \alpha$, where $a + b\alpha = c + d\alpha$ means $a = c$ and $b = d$. Define

$$(a + b\alpha) + (c + d\alpha) := (a + c) + (b + d)\alpha,$$
$$(a + b\alpha)(c + d\alpha) := (ac + bd) + (ad + bc + bd)\alpha,$$

with all arithmetic of coefficients in $\mathbb{F}_2$. (The product is what multiplying out gives if $\alpha^2$ is replaced by $1 + \alpha$.)

(a) Write out the addition and multiplication tables of $F$.

(b) Prove that $F$ is a field.

(c) Show that there is no bijection $\varphi\colon F \to \mathbb{Z}/4\mathbb{Z}$ with $\varphi(x + y) = \varphi(x) + \varphi(y)$ and $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in F$. So this four-element field is not $\mathbb{Z}/4\mathbb{Z}$ in disguise.

*Solution* (Hint 1). For (c): where must $\varphi$ send 0?

*Solution* (Hint 2). For (c): look for a non-zero element of $\mathbb{Z}/4\mathbb{Z}$ whose square is zero, and ask which element of $F$ is sent to it.

*Solution.*

(a) Using the formulas, with $1 + 1 = 0$ in $\mathbb{F}_2$:

| $+$ | $0$ | $1$ | $\alpha$ | $\beta$ |
|---|---|---|---|---|
| $0$ | $0$ | $1$ | $\alpha$ | $\beta$ |
| $1$ | $1$ | $0$ | $\beta$ | $\alpha$ |
| $\alpha$ | $\alpha$ | $\beta$ | $0$ | $1$ |
| $\beta$ | $\beta$ | $\alpha$ | $1$ | $0$ |

| $\cdot$ | $0$ | $1$ | $\alpha$ | $\beta$ |
|---|---|---|---|---|
| $0$ | $0$ | $0$ | $0$ | $0$ |
| $1$ | $0$ | $1$ | $\alpha$ | $\beta$ |
| $\alpha$ | $0$ | $\alpha$ | $\beta$ | $1$ |
| $\beta$ | $0$ | $\beta$ | $1$ | $\alpha$ |

For instance, $\alpha\alpha = (0 + 1\alpha)(0 + 1\alpha) = (0 + 1) + (0 + 0 + 1)\alpha = 1 + \alpha = \beta$, and $\alpha\beta = (0 + 1\alpha)(1 + 1\alpha) = (0 + 1) + (0 + 1 + 1)\alpha = 1$.

(b) Let $x = a + b\alpha$, $y = c + d\alpha$, $z = e + g\alpha$ with $a, b, c, d, e, g \in \mathbb{F}_2$. Since $\mathbb{F}_2$ is a field (Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$)), the coefficients obey every rule of Section 1.1, and in addition $u + u = (1 + 1)u = 0$ for every $u \in \mathbb{F}_2$.
*Addition* works on each coefficient separately, so (F1) and (F2) for $F$ are (F1) and (F2) for $\mathbb{F}_2$, applied twice; the element $0 = 0 + 0\alpha$ satisfies (F3); and $-x = x$ satisfies (F4), since $a + a = 0$ and $b + b = 0$.
*(F6).* Exchanging $(a, b)$ with $(c, d)$ leaves the product formula as it was: $ac+bd = ca+db$ and $ad + bc + bd = cb + da + db$, by (F1) and (F6) in $\mathbb{F}_2$.

*(F7).* With $1 = 1 + 0\alpha$, $x \cdot 1 = (a \cdot 1 + b \cdot 0) + (a \cdot 0 + b \cdot 1 + b \cdot 0)\alpha = a + b\alpha = x$. And $1 \neq 0$, because the first coefficients 1 and 0 of $\mathbb{F}_2$ differ.
*(F9).* Since $y + z = (c + e) + (d + g)\alpha$,

$$\begin{aligned} x(y+z) = {} & \big(a(c+e) + b(d+g)\big) \\ & + \big(a(d+g) + b(c+e) + b(d+g)\big)\alpha, \end{aligned}$$

and multiplying out the coefficients regroups this as $\big((ac + bd) + (ae + bg)\big) + \big((ad + bc + bd) + (ag + be + bg)\big)\alpha = xy + xz$.
*(F5).* Write $xy = h + k\alpha$ with $h = ac + bd$ and $k = ad + bc + bd$. Then $(xy)z = (he + kg) + (hg + ke + kg)\alpha$, and multiplying out,

$$\begin{aligned} he + kg &= ace + adg + bcg + bde + bdg, \\ hg + ke + kg &= acg + ade + adg + bce + bcg + bde, \end{aligned}$$

where the second line uses $bdg + bdg = 0$. On the other side, $yz = l + m\alpha$ with $l = ce + dg$ and $m = cg + de + dg$, and $x(yz) = (al + bm) + (am + bl + bm)\alpha$. Multiplying out gives $al + bm = ace + adg + bcg + bde + bdg$ and $am + bl + bm = acg + ade + adg + bce + bcg + bde$, again using $bdg + bdg = 0$. The coefficients agree, so $(xy)z = x(yz)$.
*(F8).* From the table, $1 \cdot 1 = 1$ and $\alpha\beta = \beta\alpha = 1$. So every non-zero element has an inverse: $1^{-1} = 1$, $\alpha^{-1} = \beta$, $\beta^{-1} = \alpha$.
This proves that $F$ is a field.

(c) Assume, for a contradiction, that $\varphi$ is such a bijection. To begin, $\varphi(0) = \varphi(0 + 0) = \varphi(0) + \varphi(0)$ in $\mathbb{Z}/4\mathbb{Z}$. Adding $-\varphi(0)$ to both sides gives $\varphi(0) = [0]$; this is additive cancellation, whose proof (Theorem 1.1.1 (b), first half) uses only (F1)–(F4), which hold in the commutative ring $\mathbb{Z}/4\mathbb{Z}$ (Number Systems: From \(\nN\) to \(\nR\), Theorem 3.4.3 (a)). Surjectivity provides $x \in F$ with $\varphi(x) = [2]$; this $x$ is not 0, because $\varphi(0) = [0] \neq [2]$. But

$$\varphi(xx) = \varphi(x)\varphi(x) = [2][2] = [4] = [0] = \varphi(0),$$

and injectivity turns this into $xx = 0$. By (b), $F$ is a field, so Theorem 1.1.1 (f) gives $x = 0$, a contradiction. Hence no such $\varphi$ exists.

**Marking scheme.** (10 points)

- 2: (a) both tables correct, with at least one product computed from the formula.
- 1: (b) addition clauses (F1)–(F4), coefficientwise, with $-x = x$.
- 1: (b) (F6) and (F7), including $1 \neq 0$.
- 2: (b) (F5) and (F9) by expanding coefficients, with $u + u = 0$ in $\mathbb{F}_2$ used and named.
- 1: (b) (F8) from the table.
- 1: (c) $\varphi(0) = [0]$, with the cancellation justified in $\mathbb{Z}/4\mathbb{Z}$.
- 2: (c) an $x \neq 0$ with $\varphi(x) = [2]$ by surjectivity, then $xx = 0$ by injectivity, contradicting the absence of zero divisors in $F$. A solution that instead takes $x$ with $\varphi(x) = [1]$ by surjectivity and uses $x + x = 0$ in $F$ (from the addition table) to get $[0] = \varphi(0) = \varphi(x + x) = [1] + [1] = [2]$ earns these 2 points; $\varphi(0) = [0]$ still earns the line above.

Fatal errors (0 for the step): "$F$ is $\mathbb{Z}/4\mathbb{Z}$ because both have four elements"; using in (c) that $\mathbb{Z}/4\mathbb{Z}$ is a field. A solution of (b) that checks (F5) and (F9) by listing all 64 triples from the tables earns the same points.

By Theorem 1.2.2, $\mathbb{Z}/4\mathbb{Z}$ is not a field, so a field with four elements cannot be found among the

rings $\mathbb{Z}/n\mathbb{Z}$; part (c) shows that $F$ is not even a disguised copy of $\mathbb{Z}/4\mathbb{Z}$. Chapter 4 builds this field again, from polynomials over $\mathbb{F}_2$, and explains where the rule $\alpha^2 = 1 + \alpha$ comes from.

**Exercise. C2: Find the error**

Here is a claim with a proof.

> **Claim.** Let $p$ be a prime. Then $\mathbb{Z}/p\mathbb{Z}$ is a field.
>
> **Proof.** (1) By Example (Arithmetic on $\mathbb{Z}/n\mathbb{Z}$ and the Field $\mathbb{F}_2$), only (F8) needs checking. (2) Let $[a] \neq [0]$; then $a \neq 0$. (3) Put $[a]^{-1} := [1/a]$. (4) Then $[a][1/a] = [a \cdot (1/a)] = [1]$. (5) So every non-zero class has an inverse, and $\mathbb{Z}/p\mathbb{Z}$ is a field.

Is the claim true? Does the proof prove it? If not, name the line that fails, the error, and give a counterexample to the step.

*Solution.* The **claim is true**: it is Theorem 1.2.2, direction ($\Leftarrow$).

The **proof is wrong**, at line (3), and line (4) inherits the fault. The classes of $\mathbb{Z}/p\mathbb{Z}$ are classes of **integers**, and "$[1/a]$" names a class only when $1/a$ is an integer, that is, when $a = 1$ or $a = -1$. The rule $[x][y] = [xy]$ of Number Systems: From \(\nN\) to \(\nR\), Definition 3.4.2 is stated for integers $x, y$ only; line (4) applies it to the non-integer $1/a$, a general "fact" that no result supplies. This is entry (N9) of Logic and Proof, Example (What is not allowed): citing a line that is not available. The proof also never uses that $p$ is prime, a sign that something is wrong: the same lines would "prove" that $\mathbb{Z}/6\mathbb{Z}$ is a field.

A counterexample to the step: $p = 5$, $a = 2$. Then $1/2 \notin \mathbb{Z}$, so $[1/2]$ is not an element of $\mathbb{Z}/5\mathbb{Z}$. (The inverse of $[2]$ exists, but it is $[3]$, which line (3) does not produce.)

A repair: since $p$ is prime and $p \nmid a$, the only positive common divisor of $a$ and $p$ is 1; Lemma 1.2.1 gives $ax + py = 1$, and then $[a][x] = [ax] = [1]$, because $p \mid ax - 1$.

**Exercise. C3**

Find all solutions of $x^2 + x + 1 = 0$ in $\mathbb{F}_2$, in $\mathbb{F}_3$ and in $\mathbb{F}_7$.

*Solution.* Each field is finite, so we can test every element; this finds all solutions.

In $\mathbb{F}_2$: the values at $x = 0, 1$ are 1 and $3 = 1$. **No solution.**

In $\mathbb{F}_3$: the values at $x = 0, 1, 2$ are 1, $3 = 0$ and $7 = 1$. The **only solution is** $x = 1$.

In $\mathbb{F}_7$: the values at $x = 0, 1, \ldots, 6$ are the remainders of $1, 3, 7, 13, 21, 31, 43$ on division by 7, namely $1, 3, 0, 6, 0, 3, 1$. The **solutions are** $x = 2$ **and** $x = 4$, since $7 = 1 \cdot 7$ and $21 = 3 \cdot 7$.

The three answers differ in number (none, one, two), although the equation looks the same. Section 1.5 returns to this equation with completing the square, which needs $1 + 1$ to be invertible: it is in $\mathbb{F}_3$ and $\mathbb{F}_7$, and it is not in $\mathbb{F}_2$.